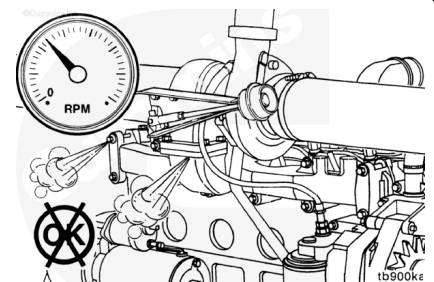


Trainings ▾

Initial Check

Inspect for exhaust leaks at the exhaust manifold, exhaust pipe, and turbocharger gasket. Check for muffler or catalyst restrictions. Leaks or restrictions will cause the turbine and impeller to operate at a lower speed and reduce the amount of air being forced into the cylinders. Again, the symptom will be excessive smoke, low-manifold pressure, and low power.

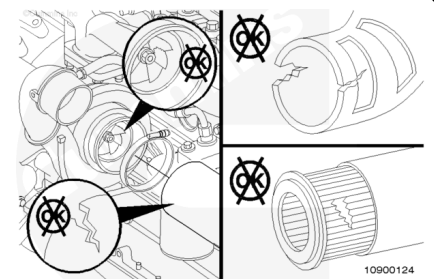


Remove the intake pipe from the turbocharger.

Inspect the turbocharger compressor impeller blades for damage.

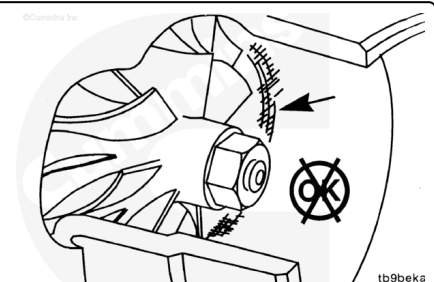
If the compressor impeller is damaged, inspect the intake piping and filter element for damage.

Replace the turbocharger if damage is found.



Failure of the internal components of the turbocharger can reduce its effectiveness and also cause excessive smoke and low power. A bearing failure can produce friction, which will slow the speed of the rotor assembly. Failed bearings can also allow the blades of the rotor assembly to rub the housings, thus reducing the rotor assembly speed.

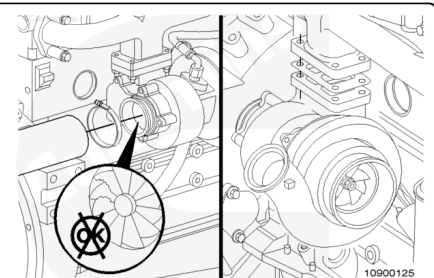
Malfunctioning turbocharger wastegate failure or miscalibration of the turbocharger wastegate can result in excessively high or low boost pressures. Low boost pressures can cause excessive smoke and low power. High boost pressures can cause major engine damage.



Remove the exhaust pipe from the turbocharger. See the Remove and Install steps in this procedure.

Inspect the turbine wheel for damage.

Replace the turbocharger, if damage is found.



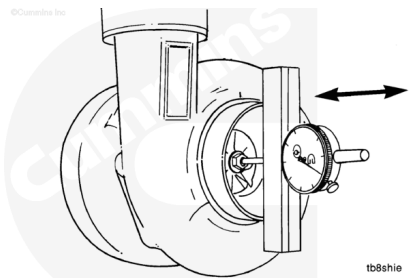
Axial Clearance Check

Measure the turbocharger shaft end clearance with the dial depth gauge, Part Number ST537.

Push the rotor assembly away from the gauge.

Set the gauge on zero.

Push the rotor assembly toward the gauge and record the data.

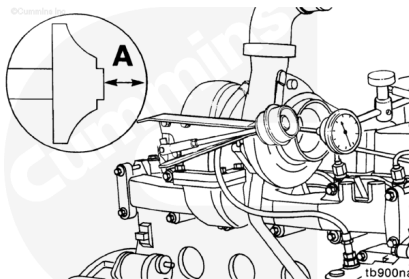


Axial Clearance (All Turbochargers except HE200WG)

mm		in
0.038	MIN	0.0015
0.093	MAX	0.0037

Axial Clearance (HE200WG)

mm		in
0.010	MIN	0.0004
0.076	MAX	0.0030



Replace the turbocharger if the clearance does **not** meet the specifications. See the Remove and Install steps in this procedure.

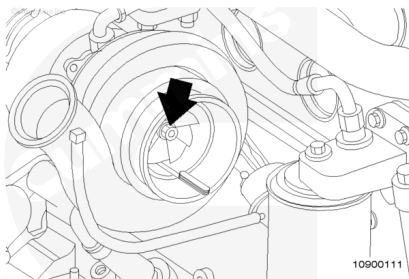
Radial Clearance Check

Measure radial clearance of the shaft.

Use a narrow blade or a wire-type feeler gauge to measure the clearance between the compressor wheel and housing.

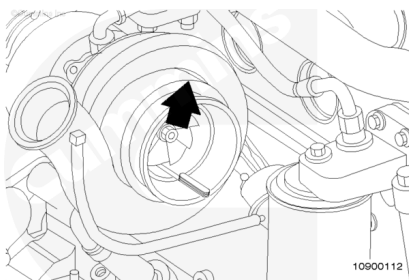
Gently push the compressor wheel toward the compressor housing and gauge.

Record this clearance.



With the feeler gauge in the same location, gently push the compressor wheel away from the compressor housing, and measure the clearance between the compressor wheel and housing.

Record this clearance.



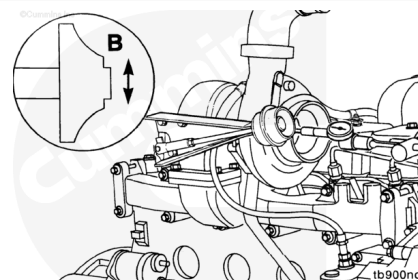
Subtract the smaller clearance from the larger clearance. This is the radial bearing clearance.

Radial Bearing Clearance (All Turbochargers except HE200WG)

mm		in
0.30	MIN	0.012
0.46	MAX	0.018

Radial Bearing Clearance (HE200WG)

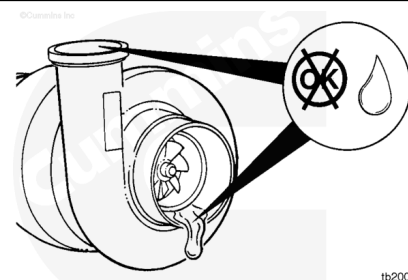
mm		in
0.394	MIN	0.0155
0.602	MAX	0.0237



The turbocharger **must** be removed for replacement if the clearances are beyond these limits. See the Remove step in this procedure.

Inspect the turbocharger compressor intake and discharge for oil.

If oil is present in the compressor intake, as well as in the discharge, check upstream of the turbocharger for the source of the oil.

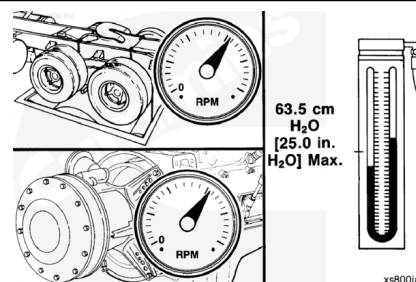


If oil is present **only** in the discharge side, install the air intake and charge air cooler piping. Refer to the OEM service manual.

Check for intake restriction. Refer to Procedure 010-031 in Section 10. (</qs3/pubsys2/xml/en/procedures/100/100-010-031.html>)

If no intake restriction is found, replace the turbocharger.

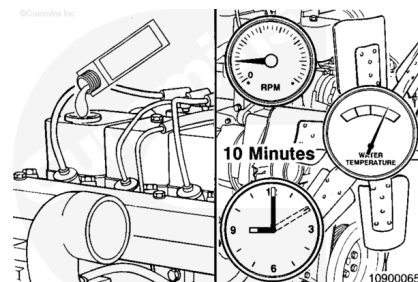
Note : If the engine experiences a turbocharger failure or any other occasion where oil is put into the charge air system, the charge air system must be inspected and cleaned. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html>)



Leak Test

Add 1 unit of fluorescent tracer, Part Number 3376891, to each 38 liters [10 gal] of engine lubricating oil.

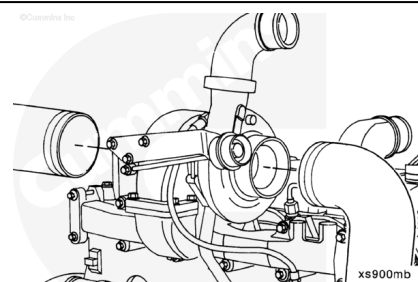
Operate the engine at low idle for 10 minutes.



Shut off the engine.

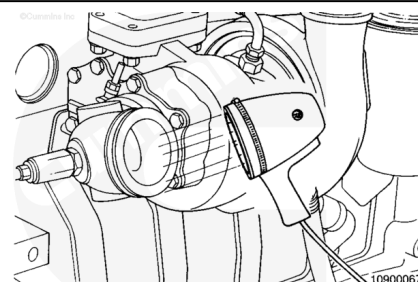
Allow the turbocharger to cool.

Remove the exhaust pipe from the turbine housing.



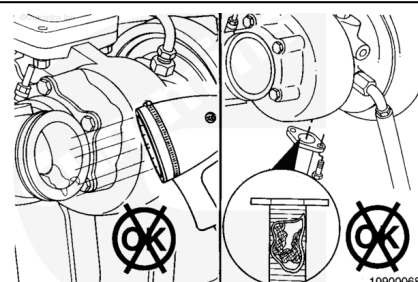
Use a high-intensity black light, Part Number 3824754 or 3824524, to inspect the turbine outlet for leaks.

A dark blue glow indicates a fuel leak. Refer to Procedure 007-044 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-044.html>) A yellow glow indicates an oil leak.

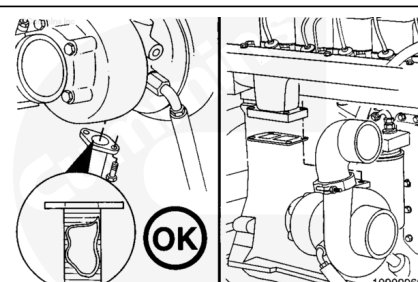


If oil is found on the turbine housing, remove the oil drain line, and check for restrictions. Clear any restrictions found, or replace damaged components as required.

If the engine is equipped with a turbocharger oil drain fitting with a screen, remove the fitting, and check to make sure the screen is clean.



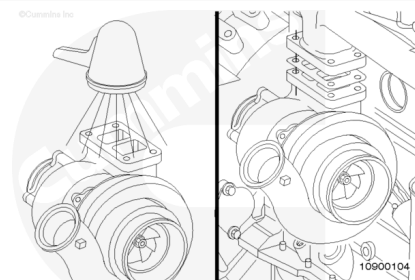
If the oil drain line is **not** restricted, remove the turbocharger. See the Remove step in this procedure.



Use a high-intensity black light, Part Number 3829754 or 3824524, to inspect the turbine inlet for leaks.

A yellow glow indicates an oil leak from the engine.

If a yellow glow is **not** seen in the turbine inlet, replace the turbocharger.



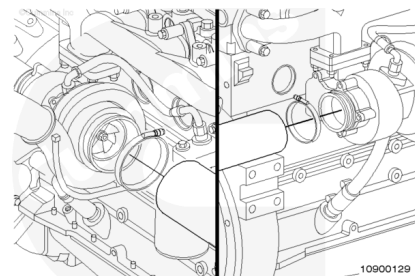
Install the intake pipe to the turbocharger compressor inlet and tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

Install the exhaust pipe to the turbocharger turbine outlet and tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

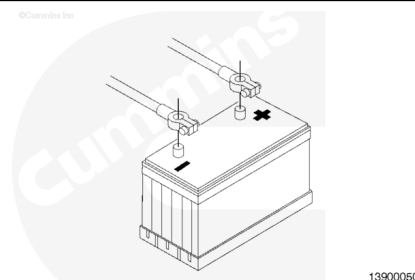
See the Install step in this procedure.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries.

- Remove air crossover tube, if equipped. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- Remove boost capsule actuator hose.
- Remove oil supply line from the turbocharger. Refer to Procedure 010-046 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-046-tr.html)



- Remove oil drain line from the turbocharger. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-045-tr.html)

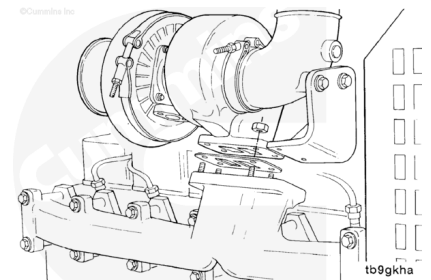
Remove

Remove the exhaust piping.

Disconnect the turbocharger compressor air inlet pipe.

Remove the turbocharger compressor outlet elbow, v-band clamp, and o-ring from the turbocharger compressor outlet.

Remove the exhaust clamp, turbocharger mounting nuts, turbocharger, and gasket.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

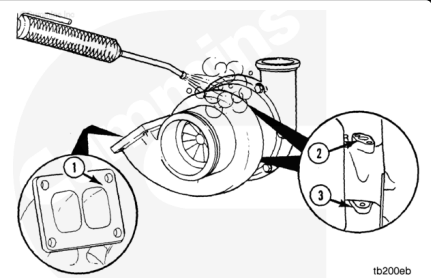
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠



Tape or plug all openings to prevent solvent or steam from damaging the oil cavities in the turbocharger.

Remove all carbon deposits and gasket material from surfaces (1), (2), and (3).

Use solvent or steam to clean the exterior of the turbocharger.
Dry with compressed air.

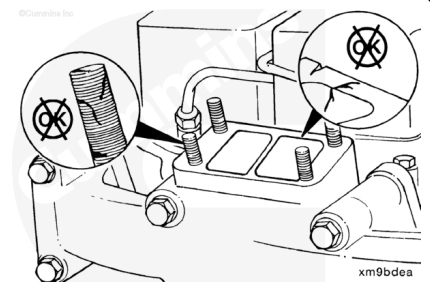
Note : If the turbocharger is **not** to be immediately replaced, cover the opening to prevent any material from falling into the manifold.

Clean the turbocharger and exhaust manifold gasket surfaces.

Inspect the turbocharger and exhaust manifold gasket surfaces and mounting studs for cracks and damage.

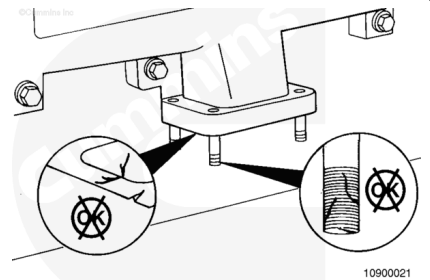
Replace the turbocharger if any cracks are found on the mounting flange surface. See the Remove and Install steps in this procedure.

Replace the exhaust manifold if any cracks are found on the mounting flange surface. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/40/40-011-007-tr.html>)



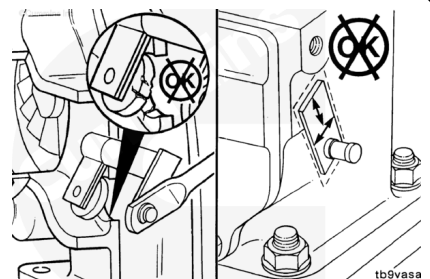
Inspect the sealing surface and mounting studs for damage.

Inspect the turbocharger.



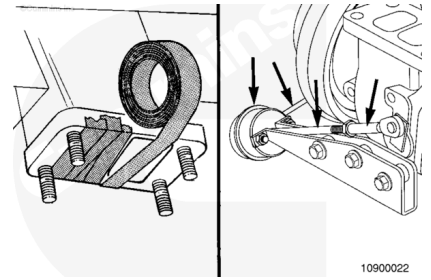
Inspect turbocharger wastegate linkage, valve shaft, module, and pressure signal line.

Check or calibrate wastegate. Refer to Procedure 010-050 in Section 10 (</qs3/pubsys2/xml/en/procedures/40/40-010-050.html>) or to the Turbocharger Shop Manual, Bulletin 3810321.



Inspect turbocharger wastegate linkage, valve shaft, module, and pressure signal line.

Inspect the turbocharger impeller blades for damage. If the impeller is damaged, inspect the intake piping and filter element for damage.

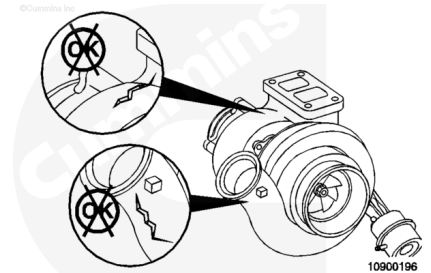


Inspect the turbine and compressor housings.

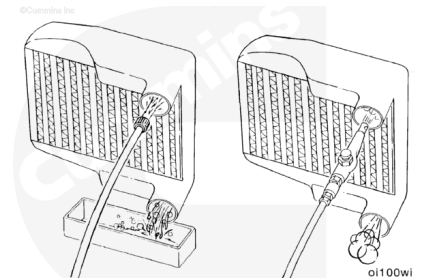
If cracks that go all the way through the outer walls are found, the turbocharger **must** be replaced.

Note : A charge air cooler failure can cause progressive damage to the turbine housing. If the turbine housing is damaged, check the charge air cooler. Refer to Procedure 010-027 in Section 10.

(/qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html)



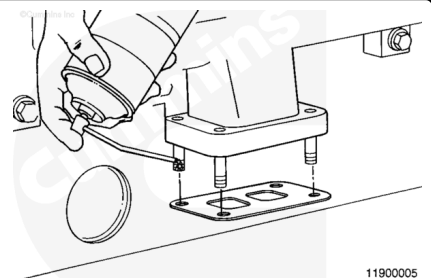
Note : If the engine experiences a turbocharger failure or any other occasion where oil is put into the charge air system, the charge air system must be inspected and cleaned. Refer to Procedure 010-027 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-027-tr.html)



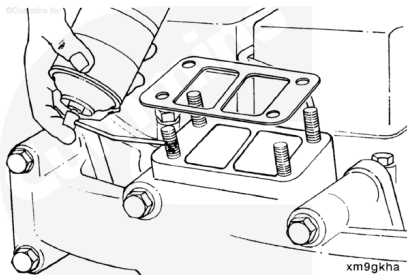
Install

Install a new gasket, and apply anti-seize compound to the mounting studs.

Note : Use **only** lead-free anti-seize compound compatible with oxygen sensors.

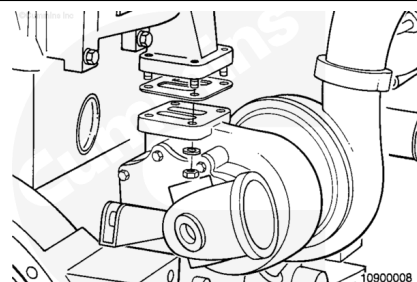


Install a new gasket, and apply anti-seize compound to the mounting studs.



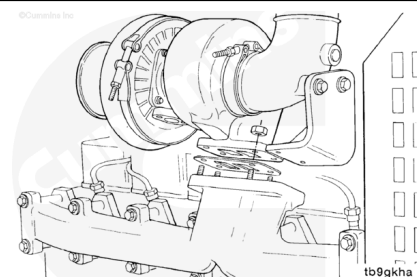
Install the turbocharger.

Torque Value: 45 n•m [33 ft-lb]



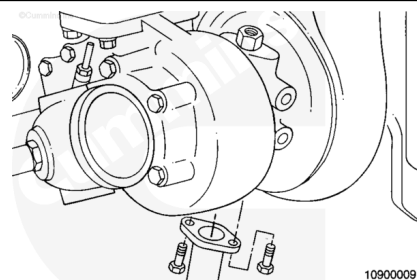
Install the turbocharger.

Torque Value: 45 n•m [33 ft-lb]

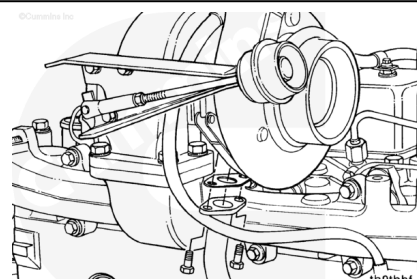


Use a new gasket and connect the oil drain tube.

Torque Value: 24 n•m [18 ft-lb]

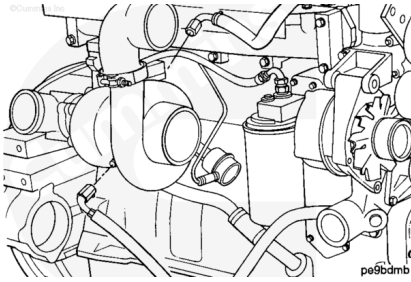


Install the turbocharger oil drain line. Refer to Procedure 010-046 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-046-tr.html)



Install the turbocharger coolant lines.

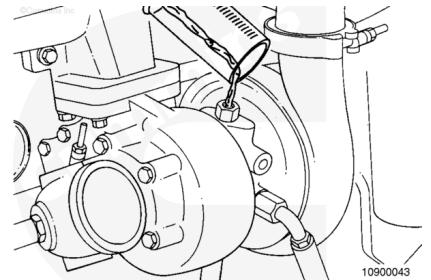
Note : Keep coolant line horizontal when tightening.



Note : New turbochargers must be prelubricated before startup.

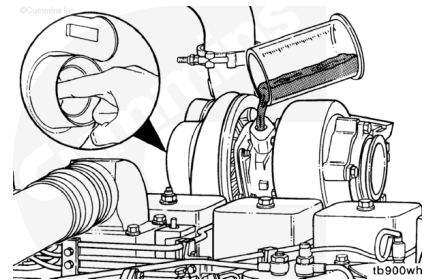
Pour 50 to 60 cc [2 to 3 oz] of clean engine oil into the oil supply fitting.

Rotate the turbine wheel to allow the oil to enter the bearing housing.



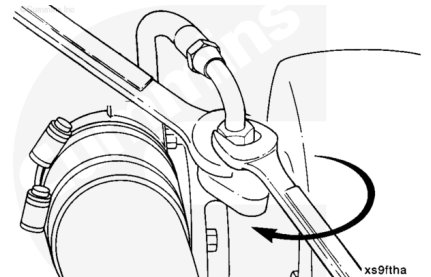
Lubricate the bearings by pouring 59 to 89 ml [2 to 3 oz] of clean 15W40 engine oil into the turbocharger oil supply line fitting.

Rotate the turbine wheel to allow oil to enter the bearing housing.

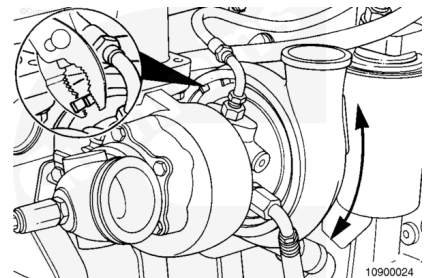


Connect the oil supply line.

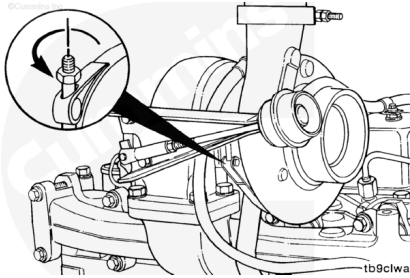
Torque Value: 35 n·m [26 ft·lb]



If required, loosen the compressor housing v-band clamp and position the housing to align with the turbocharger air outlet tube.



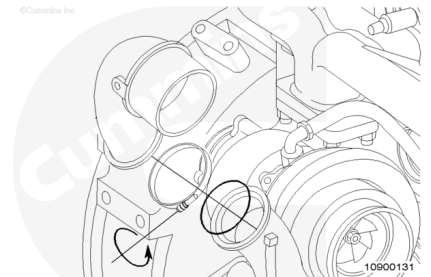
If required, loosen the compressor housing v-band clamp and position the housing to align with the turbocharger air outlet tube.



Install the turbocharger compressor outlet elbow, v-band clamp, and new o-ring seal on the turbocharger compressor discharge outlet.

Tighten the clamp.

Torque Value: 8 n•m [71 in-lb]

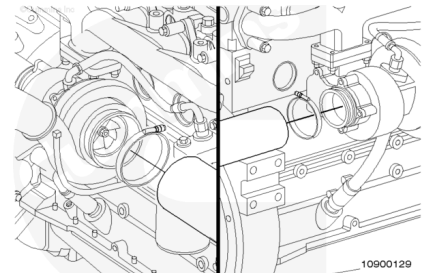


Install the intake pipe to the turbocharger compressor inlet and tighten the clamp.

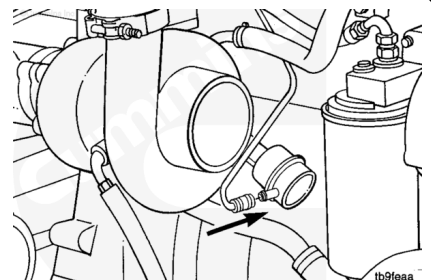
Torque Value: 8 n•m [71 in-lb]

Install the exhaust pipe to the turbocharger turbine outlet and tighten the clamp.

Torque Value: 8 n•m [71 in-lb]



Install the wastegate actuator capsule actuator hose.



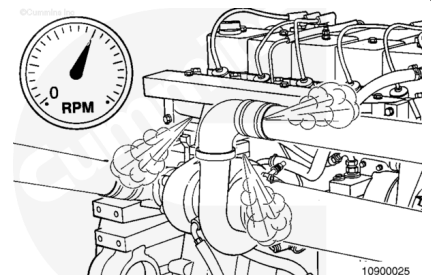
Install charged air cooler hose.

Install the air inlet and exhaust piping.

Torque Value: 8 n•m [71 in-lb]

Fill the cooling system. Refer to Procedure 008-018 in Section 8. ([/qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html](http://qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html))

Operate the engine. Check for leaks.



Finishing Steps

Install charge air cooler hose.

©Cummins Inc.

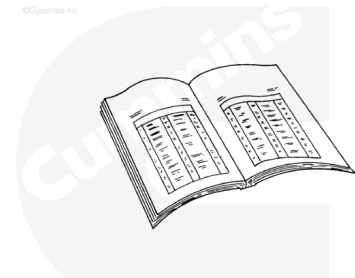
Install the air inlet and exhaust piping.

Torque Value: 8 n•m [71 in-lb]

Operate the engine and check for leaks.

- Install the turbocharger oil supply line. Refer to Procedure 010-045 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-045-tr.html)
- Install the air crossover tube, if used. Refer to Procedure 010-019 in Section 10.
(/qs3/pubsys2/xml/en/procedures/40/40-010-019-tr.html)
- Install the boost capsule actuator hose.

©Cummins Inc.

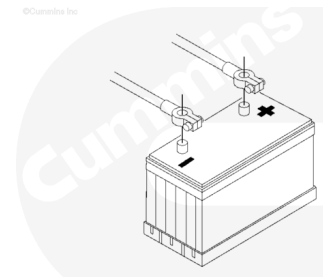


ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

©Cummins Inc.



13900050

- Connect the batteries
- Operate the engine. Check for leaks.